

Blending Neural Control Density Functions for Stabilization and Safety (ICML)

1 Comparison with Yang et al. [2024]

As requested by the reviewer hvGg, we make an comparison of our method and Yang et al. [2024]. We consider the following dynamics:

1. **Inverted Pendulum:** The dynamics of the system is given by,

$$\dot{\theta} = \dot{\theta} \quad \ddot{\theta} = \frac{MgL \sin \theta - \mu \dot{\theta}}{ML^2} + \frac{u}{ML^2} \quad (1)$$

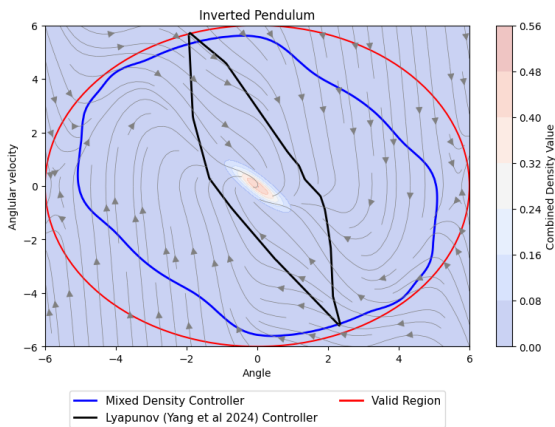
where $M = 0.15, g = 9.81, L = 0.5, \mu = 0.1$.

2. **Path following:** We consider the dynamics as reported in Yang et al. [2024],

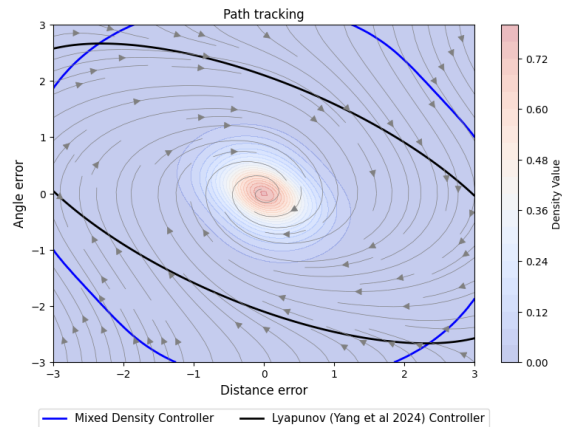
$$\dot{d}_e = v \sin \theta_e \quad \dot{\theta}_e = -\frac{v \cos \theta_e}{r - v \sin \theta_e} + \frac{v}{L} u \quad (2)$$

where $v = 2, L = 1$ and $r = 10$.

We reuse the codebase shared by Yang et al. [2024] at github to learn the Lyapunov certificate and controller for given system. We learn a Lyapunov function with hidden units [16, 16, 8] for inverted pendulum and quadratic Lyapunov function for path following. We reuse the weights from our last training for inverted pendulum. Considering the system for path tracking is different than the ones we initially trained for, we consider each network to have hidden units of 12 and 4 layers and learn new pairs of (control, density)¹.



Comparison of RoA between the blended controller system and Yang et al. [2024] for inverted pendulum system



Comparison of RoA between the blended controller system and Yang et al. [2024] for path following

¹We use the falsifier around the box to be consistent with implementation of Yang et al. [2024]

References

Lujie Yang, Hongkai Dai, Zhouxing Shi, Cho-Jui Hsieh, Russ Tedrake, and Huan Zhang. Lyapunov-stable neural control for state and output feedback: A novel formulation, 2024. URL <https://arxiv.org/abs/2404.07956>.